



Predicting upwelling dynamics in the South Sea of Java, Indonesia: A deep learning approach with ConvLSTM and 3D-CNN

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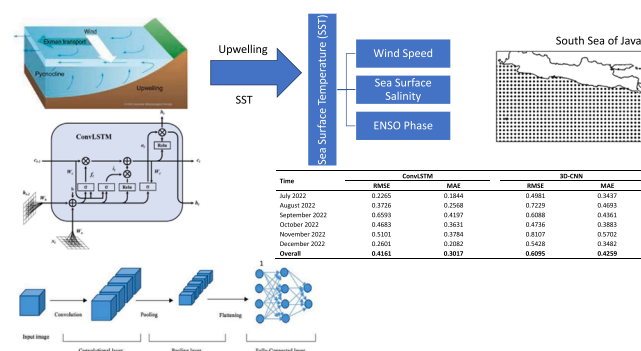
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GRAPHICAL ABSTRACT



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ABSTRACT

Oceans exhibit complex dynamics influenced by climate change, anthropogenic activities, and natural phenomena. Understanding these dynamics is critical for ensuring the sustainability of marine environments and their optimal utilization. This research aims to study and monitor upwelling phenomena in the South Sea of Java. Upwelling, the exchange of nutrient-rich, cold water from deeper layers to the surface, enhances marine biological productivity; Sea Surface Temperature (SST) serves as a key indicator for its detection. To achieve these objectives, this study employs both ConvLSTM and 3D-CNN. ConvLSTM, a deep learning architecture that integrates convolutional structures within LSTM units, effectively captures spatiotemporal dependencies in sequential data. 3D-CNN, a deep learning model extending traditional 2D convolutional neural networks, processes volumetric data, enabling the extraction of spatial features across three dimensions. Analysis reveals that ConvLSTM outperforms 3D-CNN in modeling upwelling data in the South Sea of Java. This is evidenced by lower Root Mean Square Error (RMSE) and Mean Absolute Error (MAE). The ConvLSTM method was then used for forecasting, and the results were validated with data obtained from local fishermen regarding their fishing expeditions. Visual analysis confirms that the ConvLSTM method accurately models upwelling data in the South Sea of Java with fishermen's schedules.

ConvLSTM and 3D-CNN methods were comparatively evaluated for modeling Sea Surface Temperature (SST) data, considering wind speed, sea surface salinity, and the El Niño-Southern Oscillation (ENSO) phase as influential factors.

Based on Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) values, the ConvLSTM method exhibited lower values, indicating superior performance compared to the 3D-CNN approach. Specifically, RMSE and MAE values for ConvLSTM were 0.4161 and 0.3017, respectively, while for 3D-CNN, the corresponding values were 0.6095 and 0.4259.

Upwelling data forecasting results were validated against local fishermen's schedules, with data collected in July 2022. Visual inspection confirmed alignment between the forecasted upwelling patterns and the fishermen's activity.

Specifications table

Subject area	Environmental Science
More specific subject area	Marine and Coastal Environment
Name of your method	ConvLSTM and 3D-CNN
Name and reference of original method	Snoussi, M., Tamim, A., El Fella, S., & Koutti, L. (2024). Spatiotemporal prediction of monthly coastal upwelling scenario in SST fields using deep-learning-based models. <i>IEEE Geoscience and Remote Sensing Letters</i> , 21, 1–5. Kang, L., Zhou, Z., Huang, J., & Han, W. (2022). Renal tumors segmentation in abdomen CT Images using 3D-CNN and ConvLSTM. <i>Biomedical Signal Processing and Control</i> , 72, 103,334
Resource availability	None

Background

The ocean is a vital natural resource for human life and the global ecosystem [1]. Complex ocean dynamics are influenced by various factors such as climate change, human activities, and natural phenomena [2,3]. Understanding these dynamics supports the sustainability of the marine environment and its optimal utilization [4]. One indicator of marine ecosystem health is the variation in Chlorophyll-a (Chl-a), which indicates biological productivity in the waters [5]. Research shows that Chl-a in the waters south of Java experiences seasonal variations influenced by climate phenomena such as the Indian Ocean Dipole (IOD) and El Niño-Southern Oscillation (ENSO) [6]. A similar phenomenon is also observed in upwelling intensity, where El Niño enhances upwelling, while La Niña weakens it [7]. Upwelling is a crucial phenomenon in ocean dynamics, bringing cold, nutrient-rich water from the bottom to the surface, thereby increasing biological productivity, such as Chlorophyll-a (Chl-a), and supporting marine ecosystems. Both natural and artificial upwelling play a crucial role in increasing marine productivity, such as increasing Chl-a content and macroalgae growth [8]. The South Java Current (SJC) has a variability pattern influenced by IOD and ENSO, which contribute to the upwelling dynamics in the region [9]. In addition, tropical storms such as Seroja and Odette cause sea surface cooling and salinity changes, which can affect stratification and strengthen local upwelling [10].

Previous research by Du et al. has investigated upwelling processes along oceanic eastern boundaries and their effects on marine productivity [11]. Their study employed climate models to predict complex future changes in upwelling patterns. The work of Cañadas et al. focuses on the Ediacaran ocean (635–541 million years ago) [12]. This research presents a quantitative reconstruction of primary productivity and total sedimentary organic carbon using basin-scale simulations. In the study conducted by Snoussi et al., the Moroccan Atlantic coast was examined using deep learning; however, their analysis solely incorporated Sea Surface Temperature (SST)

Table 1
Explanation of research variables.

Variable	Explanation
time	Monthly timestamp used to align all variables on a common temporal axis.
lat	Latitude of the grid cell (regular 0.25° spacing over the study domain).
lon	Longitude of the grid cell (normalized to the [−180°, 180°] convention).
sst	Sea surface temperature; target variable for upwelling intensity analyses/models.
wind_speed	10-m wind speed used to diagnose wind stress and Ekman forcing.
u10	Zonal (east–west) 10-m wind component (positive eastward).
v10	Meridional (north–south) 10-m wind component (positive northward).
chl	Chlorophyll-a concentration; proxy for biological response to upwelling.
sss	Sea surface salinity field regridded to the common 0.25° grid.
oni	NOAA's primary index for tracking the ocean part of ENSO climate pattern) aligned monthly.

as a predictive factor [13]. In the aforementioned studies, the application of deep learning methods for upwelling modeling remains limited. A significant portion of the existing research relies primarily on conventional oceanographic calculations.

This research aims to model upwelling phenomena using deep learning techniques, specifically employing Convolutional LSTM (ConvLSTM) and 3D Convolutional Neural Networks (3D-CNN). The modeling process incorporates several key oceanic physical characteristics, including the influence of the El Niño–Southern Oscillation (ENSO) on upwelling dynamics, Chlorophyll-a variations, and the impact of surface activities on temperature and salinity profiles. By comparatively assessing the performance of ConvLSTM and 3D-CNN methodologies, the research seeks to identify the most suitable approach for upwelling data modeling, with a particular focus on the South Sea of Java.

Method details

Data acquisition

To identify the magnitude of upwelling, SST was taken from NOAA's Daily Optimum Interpolation SST (OISST v2.1) at 0.25° grid resolution and daily cadence (1982–2022). The product blends satellite AVHRR retrievals with in-situ observations and sea-ice analyses via optimal interpolation to yield spatially complete, bias-adjusted fields suitable for climate analysis (<https://psl.noaa.gov/data/gridded/data.noaa.oisst.v2.html>). To quantify biological response to upwelling, MODIS-Aqua Level 3 ocean-color products for chlorophyll-a at 4 km grid spacing are used. We analyzed the Level-3 daily binned files over our study region (South Sea of Java) to form daily regional means from 2002–2022. Data were obtained from NASA's Ocean Biology Processing Group Level-3 portal (<https://oceancolor.gsfc.nasa.gov/13/>). The wind speed data was obtained from the ECMWF Reanalysis (ERA 5) archives with a spatial resolution of 0.25° x 0.25° at regular intervals (monthly dataset from 1940 to 2025). It has been found to be well-aligned and suitable for analyzing wind stress and curl in order to assess upwelling intensities (<https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels-monthly-means?tab=download>).

ENSO conditions were characterized using NOAA's Oceanic Niño Index (ONI), defined as the 3-month running mean of ERSSTv5 sea-surface temperature anomalies over the Niño-3.4 region (5°N–5°S, 120°–170°W), with event listings taken from CPC's official "Cold & Warm Episodes by Season (ONI v5)" table (1950–present) (https://origin.cpc.ncep.noaa.gov/products/analysis_monitoring/ensostuff/ONI_v5.php). ONI was analyzed to contextualize interannual variability in upwelling metrics, as ENSO modulates winds, Ekman transport, and thermocline depth in the southeast Indian Ocean, thereby influencing coastal upwelling intensity south of Java. Sea surface salinity was taken from the Copernicus Marine Service Level-4 gap-free daily analyses at 1/8° resolution, generated via a multivariate optimal interpolation that fuses satellite sea surface salinity from NASA Soil Moisture Active Passive (SMAP) and ESA's Soil Moisture Ocean Salinity (SMOS) (https://data.marine.copernicus.eu/product/MULTIOBS_GLO_PHY_S_SURFACE_MYNRT_015_013/).

A unified monthly dataset for the southern Java domain was then constructed by harmonizing time, lat, lon, sst, wind_speed, u10, v10, chl, sss, and oni onto a common 0.25° × 0.25° grid and timeline. Longitudes were normalized to the [−180°, 180°] convention and spatial coordinates restricted to the study area. Variables with native resolutions differing from the target were regridded in coastal or sparsely sampled cells. All inputs were aggregated to calendar months, and short gaps were infilled within grid cells using time-aware linear interpolation to preserve seasonal structure. An explanation of each data used can be seen in Table 1.

Data preprocessing

Satellite-derived geophysical data contain NaN (Not a Number) values over land pixels, whereas deep learning models cannot directly process NaN values during training. To address this issue, a two-stage spatial interpolation strategy was applied to fill missing values over valid sea areas only. Sea Surface Temperature (SST) data were first temporally aggregated into monthly means. The SST fields were then regridded onto a regular 0.25° × 0.25° latitude–longitude grid using bilinear interpolation, which estimates values at each target grid cell from the weighted contribution of the four nearest neighbouring source pixels, ensuring smooth spatial transitions. Grid cells that remained undefined after bilinear interpolation—primarily located in coastal regions or outside the convex hull of the original observations—were subsequently filled using nearest neighbour interpolation as a fallback method, resulting in complete

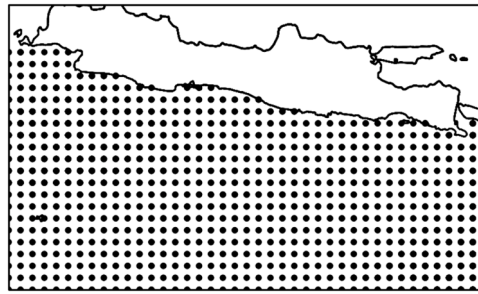


Fig. 1. Research area.

spatial coverage of SST over the analysed ocean domain.

Near-surface wind data, including zonal (u_{10}) and meridional (v_{10}) wind components, were processed using the same spatial and temporal framework to ensure consistency across variables. The wind fields were aggregated into monthly means and regridded to the same $0.25^\circ \times 0.25^\circ$ spatial resolution using bilinear interpolation. Similar to SST, residual missing values after bilinear interpolation—mainly occurring in coastal or sparsely observed grid cells—were filled using nearest neighbour interpolation. Wind speed was subsequently derived from the interpolated u_{10} and v_{10} components using the Euclidean norm, ensuring physically consistent wind magnitude estimates at each grid cell.

Large-scale climate variability associated with the El Niño–Southern Oscillation (ENSO) was represented using the Oceanic Niño Index (ONI). Monthly ONI values were temporally aligned with the gridded SST and wind datasets. As ONI is a basin-scale index without inherent spatial variability, the same monthly ONI value was uniformly assigned to all grid cells within the study domain for each corresponding month. In addition to the continuous ONI values, a simplified ENSO phase classification (El Niño, La Niña, or Neutral) was derived based on standard ONI thresholds and similarly broadcast across the spatial grid. This approach allows the model to incorporate large-scale climate forcing while preserving the spatial structure of local ocean–atmosphere variables.

Chlorophyll-*a* (CHL) concentration data derived from MODIS-Aqua Level-3 products were processed using the same temporal and spatial harmonization framework to ensure consistency with the SST and wind datasets. Daily CHL observations were first aggregated into monthly means to reduce short-term variability and data gaps caused by cloud contamination. The monthly CHL fields were then regridded onto the common $0.25^\circ \times 0.25^\circ$ latitude–longitude grid using bilinear interpolation, which computes grid-cell values from the weighted contribution of the four nearest neighbouring source pixels. Due to the presence of persistent missing values in coastal waters and regions affected by cloud cover, some grid cells remained undefined after bilinear interpolation. These remaining gaps were subsequently filled using nearest neighbour interpolation as a fallback method, assigning values from the closest valid source pixel.

Sea Surface Salinity (SSS) data were obtained from the Copernicus Marine Service Level-4 monthly products and processed using the same spatial harmonization framework as the other geophysical variables. Although the source product provides spatially complete fields through optimal interpolation, the native grid resolution differs from the target $0.25^\circ \times 0.25^\circ$ grid used in this study. Therefore, the monthly SSS fields were regridded onto the common analysis grid using bilinear interpolation, allowing smooth spatial transitions between neighbouring grid cells. To ensure full consistency across all variables and to handle occasional undefined values arising from coastal geometry or grid boundary effects, nearest neighbour interpolation was applied as a fallback method to fill any remaining missing grid cells after bilinear interpolation.

The next preprocessing step entails the land masking process, which involves the identification and elimination of land pixels from the analysis. This is due to the fact that the focal point of this study is ocean dynamics [14]. This process utilizes a 1:50 m scale Natural Earth shapefile to delineate land boundaries. In order to convert each lat-lon coordinate in the dataset into point geometry, the Python library GeoPandas is utilized. Subsequently, a spatial join is executed with land polygons to generate a flag that indicates whether a pixel is situated on land or sea. However, this rudimentary land masking approach proves inadequate to address the intricate geographical characteristics of the South Sea of Java. It is important to note that there are pixels north of Java that are technically classified as sea, yet they are not pertinent to the study of upwelling occurring south of the island.

To address this challenge, an approach referred to as "extended land masking" approach was developed using coastlines, whereby for each specified longitude, the system identifies the minimum latitude of land pixels to delineate the southern boundary of the land. The resultant function is continuous, mapping each longitude to its latitude threshold. Pixels with latitudes exceeding the specified threshold are excluded. This process produces a static binary mask (1 for south java sea, 0 for land and north java sea) with dimensions of 25×41 that remains constant throughout the study period because the coastline does not change over time [14]. For the purpose of further clarification, Fig. 1 presents a visual representation of the sea data from South Sea of Java utilized in the study subsequent to the masking process.

Despite the utilisation of interpolation in the training data, this land mask assumes a pivotal function in the model evaluation stage. In order to ensure the validity of the research, it is imperative to employ an evaluation metric that exclusively calculates errors on valid sea pixels within the designated research area. This approach ensures that land pixels and areas outside the research domain are effectively ignored, thereby ensuring that the model's performance is evaluated based on the dynamics of the southern Java Sea. Furthermore, the land mask is utilised in the upwelling detection process to identify coastal pixels (coastline) and offshore pixels (pixels to the south), as will be elucidated in the Upwelling Detection section.

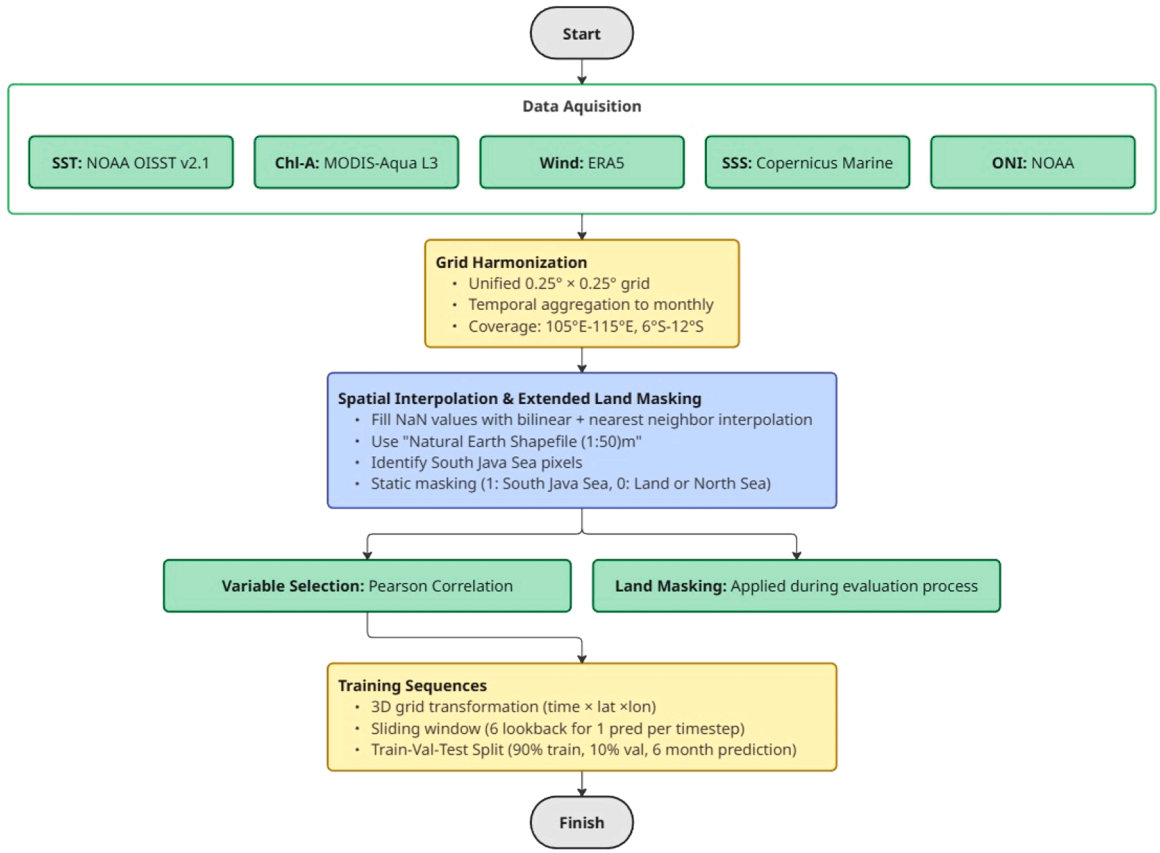


Fig. 2. Data pre-processing flowchart.

Prior to grid transformation, the most relevant predictors for SST modelling were identified by means of variable selection based on Pearson and Spearman correlation analysis. The evaluation of all variables (i.e. wind speed, u10, v10, chlorophyll-a, sea surface salinity and the oceanic Nino index) was conducted on the basis of their relationship with the target variable (SST). The selection criteria were established with a minimum threshold of moderate correlation ($|corr| \geq 0.4$) to ensure that only variables with substantial explanatory power were retained in the model [15].

The subsequent phase involves the conversion of the initial tabular dataset into a three-dimensional grid structure, a transformation necessary to align with the spatio-temporal neural network architecture [16]. The initial step in this process is the sorting of the data based on the timestamp, with the objective of ensuring the correct temporal order. The unique values extracted from the latitude and longitude are then sorted to form a regular grid, with dimensions of 25 latitude points and 41 longitude points. This grid covers the geographical area between 105°E and 115°E and 12°S and 6°S . Subsequently, a function was developed to transform each variable from a long-form format (rows $\times$ columns) to a 3D grid format (time $\times$ lat $\times$ lon). This function utilizes the pandas `pivot_table` method to reorganize the data, employing time as the index and the combination (lat, lon) as the columns. The 3D array is then initialized with NaN values to accommodate coordinates that have no data, and populated with values from the pivot table. This process is applied to all variables.

To facilitate the model's capacity to discern temporal dependencies, the data is metamorphosed into sequence pairs employing a sliding window approach [17]. The present study implements the logic with two key hyperparameters: the lookback window, which is set at six timesteps, and the prediction horizon, which is set at one timestep ahead [18]. For each valid temporal position t , the input sequence X_t is constructed by extracting six consecutive timesteps before t : The sequence is as follows: $t - 5, t - 4, t - 3, t - 2, t - 1, t$. The target output y_t is defined as the SST value at timestep $t + 1$. The sliding window process is a method that produces overlapping sequences, thereby maximizing the number of training samples from a limited time series [19]. The final configuration of the training sequences is $(183 \times 6 \times 25 \times 41 \times n)$ for the input and $(183 \times 25 \times 41 \times 1)$ for the target. Subsequently, the training sequences are divided into two subsets: a training subset constituting 90 % of the total and a validation subset comprising the remaining 10 %. This approach facilitates the process of hyperparameter tuning and the implementation of early stopping. For the test set, sequence generation is performed with the constraint that the target timestep must be within the specified test period (July-December 2022). Consequently, a proportion of the timesteps from the training period are utilized as a lookback window to predict the initial timestep of the test set. This approach is founded on a realistic premise, as in operational forecasting, historical observations are consistently available for the purpose of making predictions for the future [20].

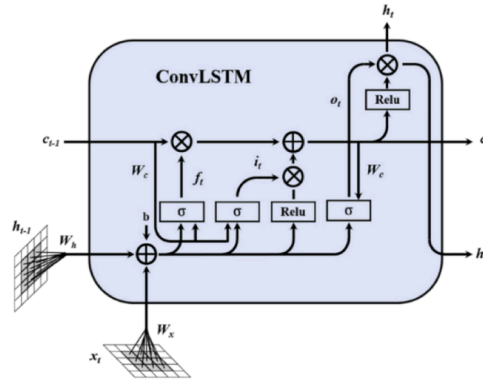


Fig. 3. ConvLSTM Cell architecture [25].

In order to elucidate the complete data preprocessing sequence, from the acquisition of satellite data to the formation of training sequence, Fig. 2 presents a comprehensive flowchart that summarises all stages from the acquisition of raw data to the formation of sequence for model training.

ConvLSTM

Convolutional Long Short-Term Memory (ConvLSTM) is an architecture specifically designed to process spatio-temporal data, integrating the capabilities of Convolutional Neural Networks (CNN) in spatial feature extraction with the temporal modeling capabilities of Long Short-Term Memory (LSTM) [21]. The rationale for employing ConvLSTM for SST prediction stems from the intrinsic characteristics of oceanographic dynamics. SST exhibits not only a complex spatial structure but also a non-linear temporal evolution with memory effects [22,23]. Current sea conditions are strongly influenced by conditions from previous weeks or months through mechanisms such as heat accumulation, advection, and mixing.

The ConvLSTM architecture successfully addresses the limitations of fully-connected LSTMs, which lose spatial structure information during the process of flattening input into vectors. Additionally, it overcomes the limitations of standard CNNs, which lack an explicit temporal memory mechanism. By replacing matrix multiplication in traditional LSTMs with convolution operations, ConvLSTMs preserve the spatial topology of data while performing temporal reasoning [24]. The ConvLSTM cell at each timestep processes spatio-temporal input and maintains two state tensors: the hidden state and the cell state. The operations in one ConvLSTM cell are visualized in Fig. 3.

Each gate in ConvLSTM has a specific role in controlling the flow of information [26]. The Input Gate establishes the extent to which novel information derived from the present input and antecedent hidden states will be incorporated into the cell state. The sigmoid output value functions as a "valve" that modulates the admission of candidate cell states. The Forget Gate is responsible for determining which information from the previous cell state should be retained or forgotten. This mechanism is critical for avoiding memory saturation and enabling the model to "reset" its memory when patterns change. The Output Gate is responsible for regulating the extent to which information from the current cell state is revealed as output. This gate functions as a filter, determining which aspects of internal memory are relevant to the prediction task. Candidate Cell State is a proposed update for the cell state. This quantity represents "new information" extracted from the current input and the previous hidden state, which is filtered by the input gate.

3D-CNN

The 3D Convolutional Neural Network (3D-CNN) employs a distinctly divergent approach to spatio-temporal learning by conceptualizing time as an ancillary spatial dimension, analogous to latitude and longitude [27,28]. In contrast to the iterative processing of sequences characteristic of recurrent networks, 3D-CNN employs convolutional operations concurrently across all three spatial-temporal dimensions. This approach enables the extraction of spatio-temporal features within a single forward pass, a process that is both expeditious and efficient [29]. The impetus for employing 3D-CNN for SST prediction stems from the observation that numerous oceanographic phenomena exhibit spatio-temporal structures that can be conceptualized as "volumes" in space-time. A notable example is the evolution of upwelling, which can be visualized as a 3D object that moves and changes shape in dimensions (time $\times$ lat $\times$ lon) [27]. Three-dimensional convolution is a process that has the capacity to capture such patterns with kernels that simultaneously consider spatial neighbors and temporal neighbors [30]. The fundamental 3D convolution operation is defined as Eq. (1):

$$Y_{[c,d,e,g]} = \sum_h \sum_{k=0}^N \sum_{i=0}^M \sum_{j=0}^M W_{[k,i,j,f,g]} \cdot X_{[c+i,d+j,e+k,n]} + b_g \quad (1)$$

where X is the input tensor with shape $(T \times H \times W \times C)$, y is the output tensor with shape $(T^l \times H^l \times W^l \times G)$, W , is the 3D con-

volutional kernel with shape $(k_t \times k_h \times k_w \times F \times G)$, c, d, e are the indices for time, height (latitude), width (longitude), k, i, j are offsets in the kernel, ranging from $-\frac{k}{2}$ to $\frac{k}{2}$ for same padding, f is the input channel index, g is the output filter index, and b_g is the bias for filter g .

Metric evaluation

In the context of spatio-temporal SST prediction in the waters south of Java, the utilization of standard evaluation metrics, such as Mean Squared Error (MSE) or Root Mean Squared Error (RMSE), confronts a fundamental challenge: the prediction grid encompasses land pixels and areas outside the research domain that lack valid SST values. Conventional metrics applied directly to the entire grid will result in evaluation signals from land pixels, which are essentially noise as these pixels have been filled through spatial interpolation and are not the focus of this study. This will cause the evaluation to reflect values on land pixels, thereby diluting the assessment of model performance on the actual research area and potentially contaminating the interpretation of actual oceanographic patterns [31].

To address this issue, the present study implements a custom masked evaluation metric that explicitly integrates the land-sea mask into the error calculation. This approach ensures that only valid sea pixels within the South Java Sea research area contribute to the metric value calculation, thereby enabling the evaluation process to focus exclusively on assessing sea dynamics performance [32]. The primary evaluation metric employed is Masked Root Mean Square Error (Masked RMSE), which is mathematically defined as Eq. (2):

$$M_{RMSE} = \sqrt{\frac{\sum_i \sum_j \left[(y_{[i,j]} - \hat{y}_{[i,j]})^2 \times \text{mask}_{[i,j]} \right]}{\sum_i \sum_j \text{mask}_{[i,j]} + \varepsilon}} \quad (2)$$

As an additional metric for model performance, the Masked Mean Absolute Error (Masked MAE) is implemented with Eq. (3):

$$M_{MAE} = \frac{\sum_i \sum_j \left(|y_{[i,j]} - \hat{y}_{[i,j]}| \times \text{mask}_{[i,j]} \right)}{\sum_i \sum_j \text{mask}_{[i,j]} + \varepsilon} \quad (3)$$

where $y_{[i,j]}$ is the actual observed SST value at pixel (i, j) , $\hat{y}_{[i,j]}$ is the model prediction SST value at pixel (i, j) , $\text{mask}_{[i,j]}$ is a binary mask with a value of 1 for south sea pixels and 0 for land and north sea pixels, i, j are indices for the latitude and longitude dimensions, and ε is a small constant for numerical stability (to prevent division by zero). Element-wise multiplication with $\text{mask}_{[i,j]}$ effectively "turns off" the contribution from land and non-research area pixels, because any squared error on pixels with $\text{mask}_{[i,j]} = 0$ will be multiplied by 0 and will not contribute to the summation. Normalization is achieved through the division of the total squared error by the number of valid sea pixels in the research area, as opposed to the total number of pixels in the grid. This ensures that the magnitude of the metric is proportional to the average error per in the research area sea pixel, unaffected by the proportion of land or interpolated areas in the grid.

Within the framework of SST prediction, both metrics offer complementary information. RMSE indicates typical magnitude error with an emphasis on large errors, while MAE provides a more robust picture of median error. Both Masked RMSE and Masked MAE are employed as evaluation metrics to comprehensively assess model performance on the actual oceanographic dynamics of the South Java Sea. This approach ensures predictions on ocean pixels demonstrate realistic spatial coherence and temporal evolution, suggesting that the model has focused on learning oceanic patterns [33].

Upwelling detection

The detection of upwelling utilizes a thermal gradient-based approach between coastal and offshore zones, leveraging the oceanographic characteristics of upwelling, defined as the phenomenon where cold water from the depths rises to the surface in coastal zones. The identification of pixels along the coastline is conducted in a systematic manner for each longitude within the grid. The algorithm first filters only sea pixels, then sorts the pixels based on descending latitude (north to south). The initial pixel in this sequence is designated as the "coastal pixel," which corresponds to the maritime point that is nearest to the coastline of Java. Subsequently, the "offshore pixel," situated six pixels south of the coastal pixel, is identified. The selection of six pixels is based on the consideration that this distance is sufficient to capture the characteristic thermal gradient of upwelling without exceeding the spatial extent of the upwelling influence zone [34]. This process yields 41 pairs of coastal-offshore points dispersed along the southern coast of Java (longitude 105°E-115°E).

Subsequently, the temperature difference computation calculates the thermal gradient for each coastal-offshore pair at each timestep. For both actual and predicted data, the surface temperature (SST) is extracted from coastal and offshore pixels, the difference between these values is calculated by Eq. (4):

$$UI_{sst} = SST_{coastal} - SST_{offshore} \quad (4)$$

A negative UI_{sst} value indicates that the temperature of coastal water is lower than the temperature of offshore water, which is an indication of active upwelling. A positive or zero UI_{sst} value indicates non-upwelling or downwelling conditions [35]. Subsequently,

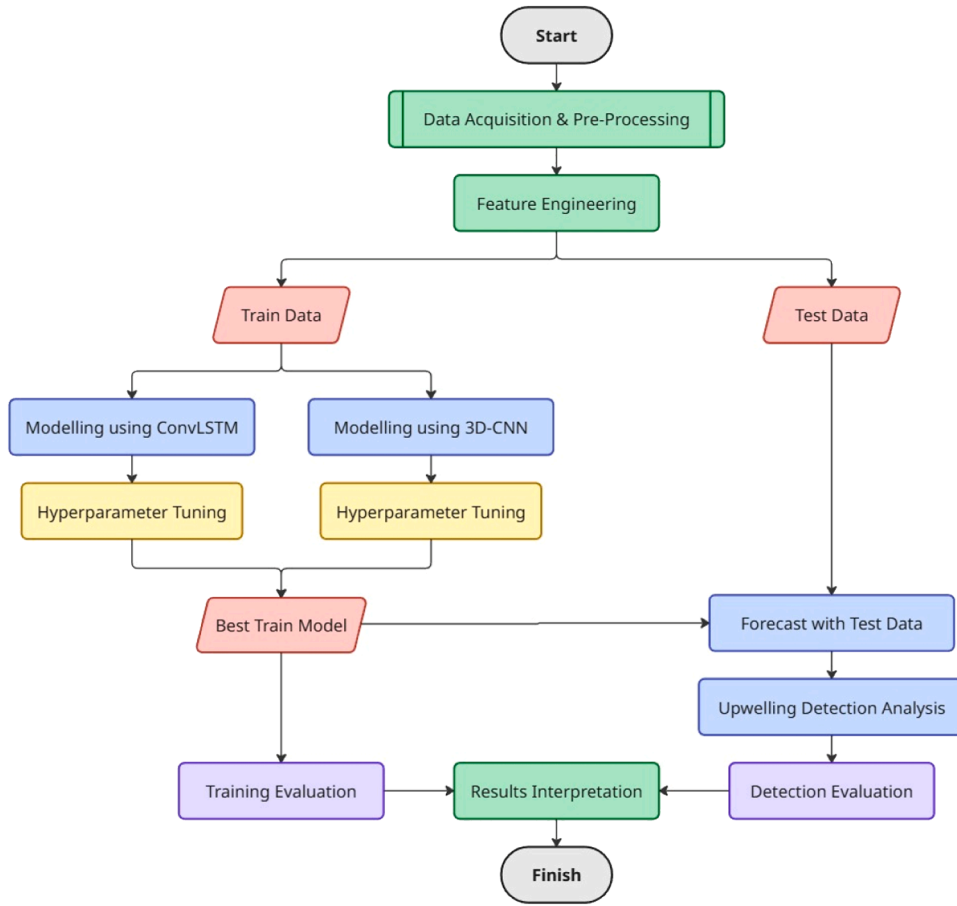


Fig. 4. Research flowchart.

binary classification is implemented through the application of a straightforward threshold: an upwelling event is identified if and only if the temperature difference (UI_{sst}) is negative. This classification is executed for both actual observations and model predictions, with each performed independently. The calculation of detection accuracy is derived from the percentage agreement between the observed upwelling and the predicted of upwelling, this formula can be seen as Eq. (5):

$$Accuracy = \frac{\sum_{i=1}^n (UI_{sst} = \widehat{UI_{sst}})}{n} \times 100\% \quad (5)$$

This metric is calculated at the timestep level to analyze temporal variability in detection capability, and aggregated as overall accuracy across all test timesteps and coastal points. The model is evaluated on a total of 246 binary classification decisions, which are derived from 41 coastal points and 6 test timesteps.

Research structure

The research process was meticulously structured to guarantee reproducibility and scientific rigor in the development of SST prediction models and upwelling detection. Fig. 4 illustrates the main stages from data acquisition to result interpretation, including the parallel development of two deep learning architectures (ConvLSTM and 3D-CNN) and a comprehensive evaluation of prediction performance and upwelling detection capabilities.

Method validation

Regridding and interpolation process

In this study, data was obtained from various satellite sources with different spatial resolutions. These discrepancies in resolution gave rise to challenges in the integration of data, as deep learning models necessitate input that is characterised by consistent grid

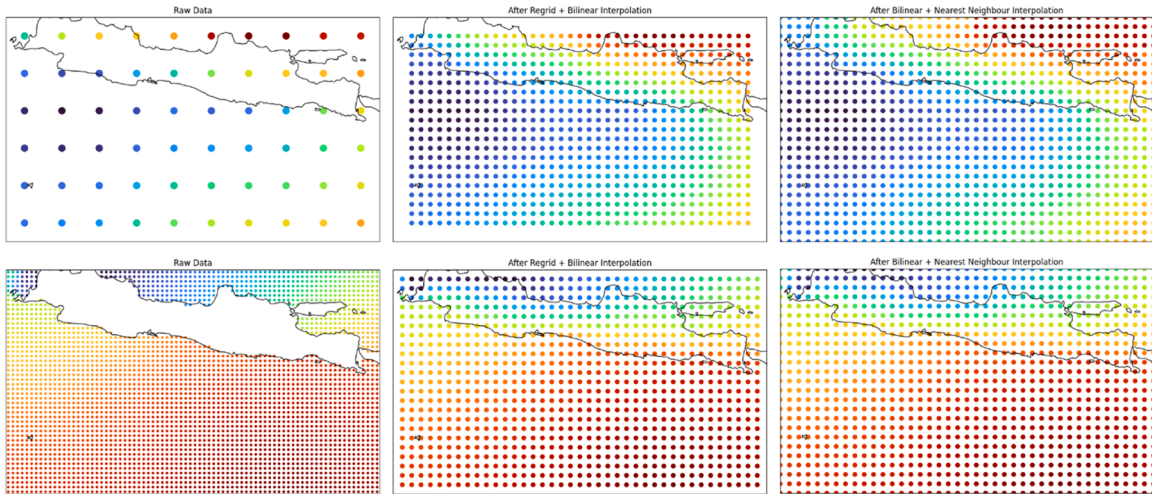


Fig. 5. Spatial regridding and two-stage interpolation process.

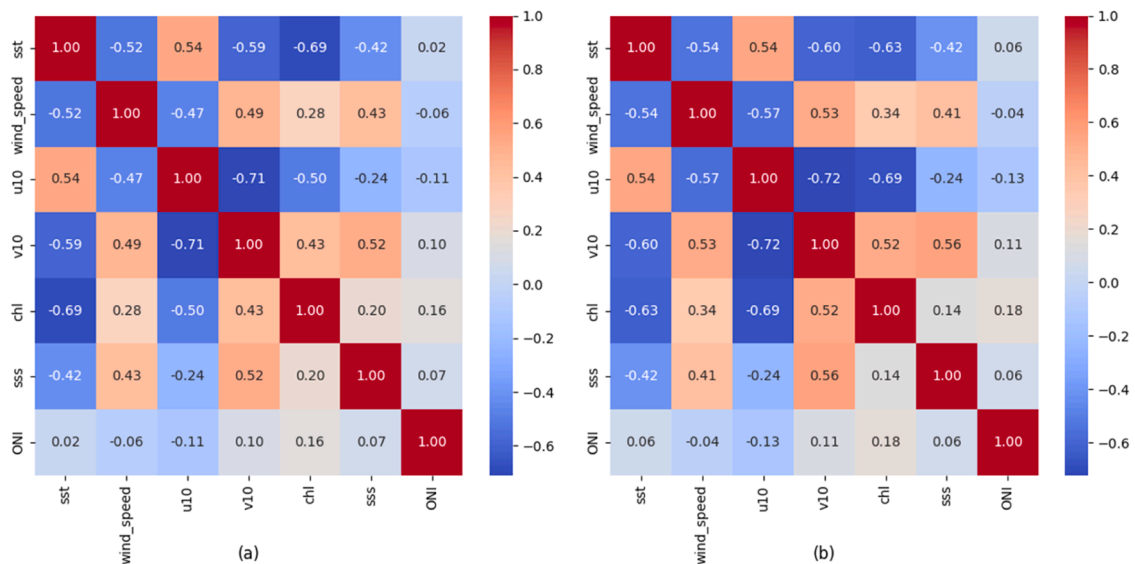


Fig. 6. Correlation matrix plot (a) Pearson's correlation (b) Spearman's correlation.

dimensions. Furthermore, satellite data contained NaN (Not a Number) values in land pixels and some unobserved sea areas, which could not be processed directly by the ConvLSTM and 3D-CNN architectures. In order to surmount this issue, a process of regridding and spatial interpolation was implemented in order to align all variables to a uniform grid with a resolution of $0.25^\circ \times 0.25^\circ$ and to fill in the missing values. Fig. 5 presents a visual representation of the regridding and interpolation process under two distinct data conditions: data exhibiting a coarser resolution than the target (top row) and data demonstrating a finer resolution than the target (bottom row).

As demonstrated in Fig. 5, the process of regridding and interpolation has a significant impact on the quality of the data. The initial column presents the raw data, characterised by varying attributes. The uppermost row exhibits data points with substantial interpoint distances, greater than 0.25° , while the lowermost row displays data points with closer interpoint distances. The second column presents the results subsequent to regridding and bilinear interpolation, where the grid has been standardised to $0.25^\circ \times 0.25^\circ$ and missing values have been substituted. However, it is important to note that bilinear interpolation alone is not sufficient to fill all pixels, especially in coastal areas or cells that are rarely observed. Consequently, the third column presents the ultimate outcomes following the implementation of nearest neighbour interpolation as a backup procedure, thereby guaranteeing that no NaN values persist within the dataset.

The amalgamation of these processes engenders a systematic and comprehensive grid, characterised by seamless value transitions in bilinearly interpolated regions and the preservation of the nearest values in inaccessible areas. This approach ensures that all

research variables possess consistent spatial dimensions (25×41 grid) and are devoid of NaN values, thereby enabling their processing by deep learning models without the occurrence of bias from artificial value substitutions such as zero. Consequently, this regridding and interpolation process becomes a critical stage in data preprocessing to ensure the integrity and consistency of model inputs.

Variable selection

Prior to grid transformation and temporal sequence formation, variable selection was performed to identify the most relevant predictors in SST modelling. The variable selection employed correlation analysis to evaluate the strength of linear and monotonic relationships between the candidate variables and the target variable (SST). Two correlation methods were utilised in a complementary manner: Pearson's correlation coefficient is employed to measure linear relationships, while Spearman's correlation coefficient is utilised to capture monotonic relationships that may be non-linear. The selection criteria were established with a minimum correlation threshold ($|\text{corr}| \geq 0.4$), ensuring that only variables with substantial explanatory power were retained in the model. This approach aligns with the established guidelines in oceanographic modelling. Fig. 6 illustrates the correlation matrix using two complementary methods: (a) Pearson's correlation for linear relationships and (b) Spearman's correlation for monotonic relationships, both of which are used to evaluate the strength of association between candidate variables and SST.

As illustrated in Fig. 6 (a), the Pearson correlation matrix demonstrates a linear relationship between the variables. The analysis results indicate that wind speed ($r = -0.52$), u10 ($r = 0.54$), v10 ($r = -0.59$), chl ($r = -0.69$), and sss ($r = -0.42$) all have correlations that exceed the set threshold, indicating a significant association with SST dynamics in the Southern Java Sea. The negative correlations between SST and wind speed, v10, and chl are indicative of the physical characteristics of upwelling, whereby increased wind speed and strong meridional components (southeasterly winds) drive offshore Ekman transport, bringing nutrient-rich cold water to the surface and increasing biological productivity (marked by an increase in chlorophyll-a). The positive correlation of u10 with SST indicates that stronger zonal wind components towards the east tend to be associated with warmer conditions, possibly related to seasonal monsoon variability. Conversely, the negative correlation between sss and SST indicates that upwelling processes contribute to the influx of saline water masses from the subsurface layer.

In order to validate the robustness of this relationship, Spearman's correlation analysis was also performed in order to capture monotonic relationships that may not be entirely linear. As demonstrated in Fig. 6 (b), the outcomes of Spearman's correlation consistently corroborate the findings of Pearson's correlation. Key variables demonstrate comparable Spearman's correlations: wind_speed ($\rho = -0.54$), u10 ($\rho = 0.54$), v10 ($\rho = -0.60$), chl ($\rho = -0.63$), and sss ($\rho = -0.42$), with all values exceeding the threshold. The strong and stable relationship between these variables and SST is confirmed by the consistency between these two correlation methods, both in a linear and monotonic context. The similarity in the magnitude of the correlations between Pearson and Spearman also indicates that this relationship is largely linear without significant distortion from outliers or extreme non-linearity [36].

Conversely, the Oceanic Niño Index (ONI), although recognised as playing an important role in large-scale climate variability, shows a very weak correlation with local SST in the Southern Java Sea. Pearson's correlation coefficient ($r = 0.02$) and Spearman's rank-order correlation coefficient ($\rho = 0.06$) both demonstrate values significantly below the established threshold, thereby indicating that the direct influence of El Niño Southern Oscillation (ENSO) on monthly SST in this region is minimal or mediated through other variables that have been incorporated into the model, such as wind patterns and salinity. Consequently, ONI was excluded from the final predictor set in order to reduce model complexity and prevent the introduction of noise that could reduce prediction accuracy. Accordingly, the variables selected for incorporation into the spatiotemporal modelling framework encompass wind_speed, u10, v10, chl, and sss as independent predictors, in conjunction with SST as the designated target variable.

Data, model, and tuning preparations

The raw dataset was transformed into a structured, three-dimensional (3D) grid format to accommodate the spatial-temporal nature of the ConvLSTM architecture. First, the data were sorted chronologically, and then unique latitude-longitude coordinates were identified to establish a regular grid structure with 25×41 spatial dimensions (latitude $\times$ longitude). Each oceanographic variable like sea surface temperature (SST), wind speed, u10 and v10 wind components, chlorophyll-a concentration, and sea surface salinity was combined into a 4D tensor with dimensions of (time $\times$ $25 \times 41 \times 1$ channels) for the target variable (SST) and (time $\times$ $25 \times 41 \times 5$ channels) for the predictor variable.

The dataset was temporally split into training and testing sets. The last six timesteps were reserved for testing, and all preceding timesteps were allocated to training. This temporal split strategy prevents data leakage and simulates real-world forecasting scenarios where future data is unavailable during model development. Each input sequence consists of six consecutive timesteps (lookback window), with the target being the SST field one timestep ahead (prediction horizon).

The ConvLSTM model captures spatio-temporal patterns in ocean dynamics by combining spatial convolution and temporal memory capabilities. Its architecture consists of the following components: (1) the ConvLSTM2D layer, which receives input in the form of ($6 \times 25 \times 41 \times 6$) and processes each timestep sequentially; (2) batch normalization, which normalizes activation distributions, accelerates convergence, and reduces sensitivity to weight initialization; (3) a ReLU activation, which introduces additional nonlinearity and improves the model's capacity to learn complex patterns; (4) a dropout layer, which serves as a form of regularization by randomly "turning off" a number of neurons during training; and (5) a Conv2D output layer, which uses 1×1 kernel functions as a fully connected layer per pixel and transforms feature maps from the ConvLSTM layer (which has dimensions of $61 \times 101 \times \text{filters}$) into final SST predictions with dimensions of ($61 \times 101 \times 1$).

The developed 3D-CNN model uses the following structure: (1) The input layer receives a tensor in the form of ($6 \times 6 \times 101 \times 5$);

Table 2
Architecture structural difference between ConvLSTM and 3D-CNN models.

ConvLSTM	3D-CNN
Input Layer (6, 25, 41, 5)	Input Layer (6, 25, 41, 5)
ConvLSTM2D (kernel 2D, processes sequentially) (25, 41, filters)	Conv3D (kernel 3D, processes simultaneously) (6, 25, 41, filters)
BatchNormalization	BatchNormalization
ReLU Activation	ReLU Activation
Dropout	Dropout
Conv2D (1, kernel 1×1)	Conv3D (1, kernel $1 \times 1 \times 1$) (6, 25, 41, 1)
Output (25 latitude, 41 longitude, 1 layer)	Lambda Layer (slice timestep -1) Output (25 latitude, 41 longitude, 1 layer)

Table 3
Comparison between five best ConvLSTM and 3D-CNN models.

Rank	ConvLSTM					3D-CNN				
	Filters	Kernel	Dropout	LR	Val RMSE	Filters	Kernel	Dropout	LR	Val RMSE
1	32	5	0.4	0.001	0.376	256	5	0.0	0.001	0.615
2	128	3	0.4	0.001	0.386	64	5	0.0	0.001	0.622
3	128	5	0.0	0.001	0.387	128	5	0.2	0.001	0.626
4	32	3	0.4	0.001	0.390	256	3	0.4	0.001	0.629
5	256	3	0.2	0.001	0.391	128	3	0.0	0.001	0.632

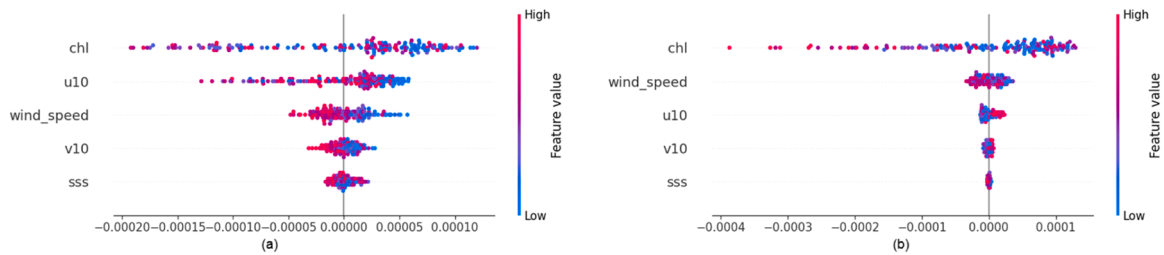


Fig. 7. Correlation matrix plot (a) Pearson's correlation (b) Spearman's correlation.

(2) the first convolutional 3D layer with an adjustable kernel size ($k \times k \times k$), where k applies to the time, latitude, and longitude dimensions simultaneously; (3) batch normalization for training stabilization; (4) dropout for regularization; and (5) a second convolutional 3D layer with a $1 \times 1 \times 1$ kernel and a single output channel to reduce the feature maps into SST predictions. This layer uses ReLU activation and same padding to maintain spatial dimensions. See Table 2 for a clarification of the structural differences between the ConvLSTM and 3D-CNN models.

To ensure a fair comparison between the two architectures, grid search was performed with an identical parameter space: number of filters (32, 64, 128, 256), kernel size (3, 5), dropout rate (0.0–0.5 with a step of 0.1), and learning rate ($1e-3$). Both the ConvLSTM and the 3D-CNN use identical custom loss functions to handle land masking. In this case, the mask is a binary array (1 for sea and 0 for land) with dimensions 25 by 41. These loss functions are implemented as TensorFlow custom functions that accept constant tensors, ensuring that only sea pixels contribute to gradient descent.

The training configuration for both models had identical settings: The Adam optimizer is used with a tuned learning rate, a maximum of 150 epochs, a batch size of 16, and early stopping with a patience of 10 epochs based on validation loss. This approach allows for a fair comparison between the two architectures, ensuring that performance differences stem from differences in temporal processing mechanisms rather than hyperparameters or training strategies.

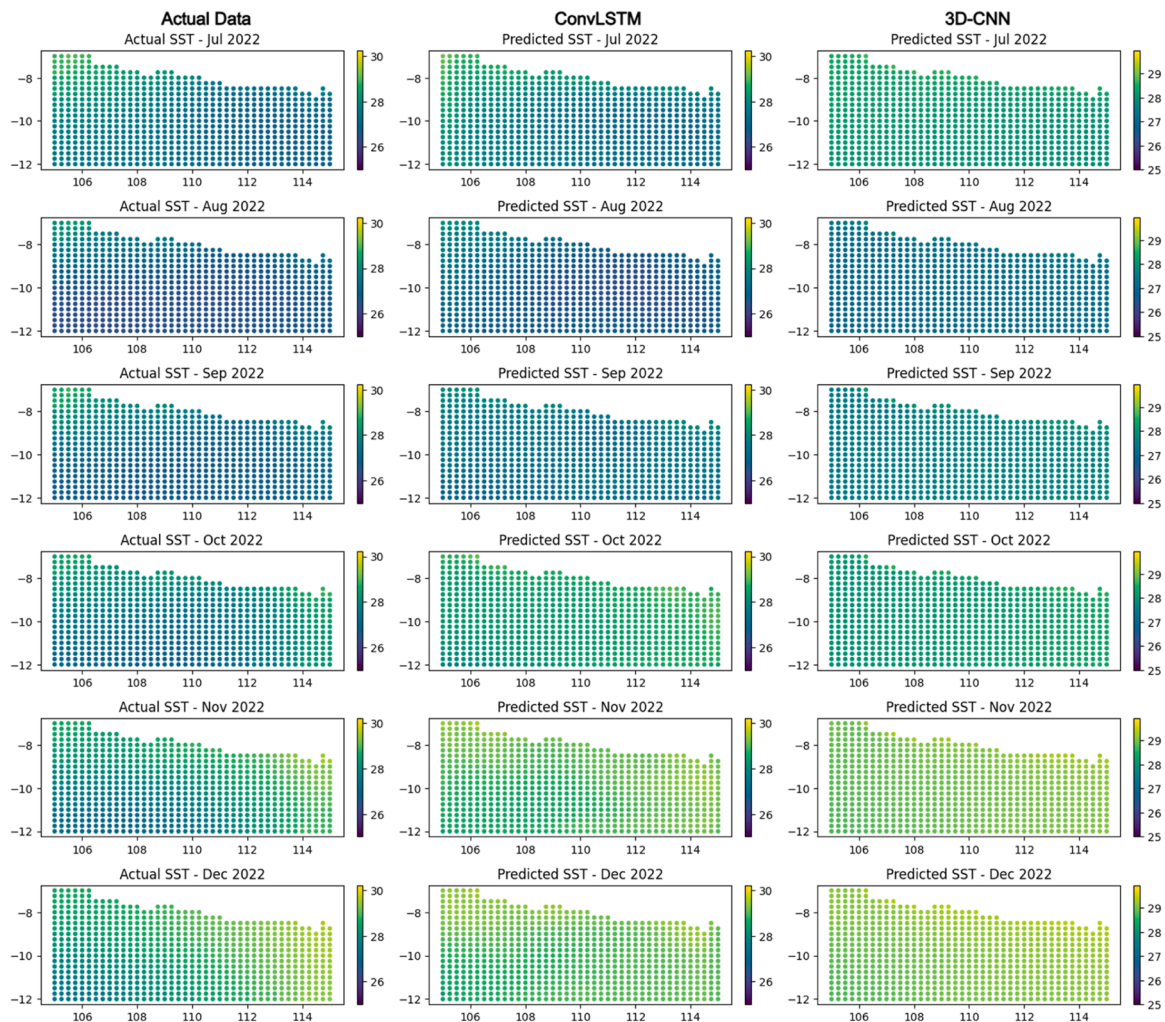
Model training evaluation

We used Keras Tuner GridSearch to tune the hyperparameters of the ConvLSTM and 3D-CNN models, running a total of 40 trials with different parameter combinations. The ConvLSTM tuning process took 32 min and 58 s, whereas the 3D-CNN process took only 16 min and 44 s. The difference in computation time reflects the ConvLSTM's sequential processing of timesteps using LSTM gates, as opposed to the 3D-CNN's ability to process the entire temporal window in parallel. ConvLSTM required an average of approximately 50 s per trial, while 3D-CNN required only 25 s, demonstrating 3D-CNN's computational efficiency, as it is 2 times faster. Table 3 provides a further comparison of the five best models between the two models.

Table 4

Evaluation result between the best ConvLSTM and 3D-CNN models.

Time	ConvLSTM		3D-CNN	
	RMSE	MAE	RMSE	MAE
July 2022	0.2676	0.2120	0.7083	0.6311
August 2022	0.3299	0.2696	0.5805	0.5301
September 2022	0.3736	0.3217	0.6887	0.6261
October 2022	0.6738	0.5746	0.6506	0.5485
November 2022	0.7265	0.6262	0.8797	0.7337
December 2022	0.4899	0.4023	0.4416	0.4416
Overall	0.4768	0.4010	0.6750	0.5851

**Fig. 8.** Actual vs Predicted SST prediction comparison from July – December 2022.

The tuning results show that the optimal configuration for ConvLSTM is 32 filters, a 5×5 kernel size, a 0.4 dropout rate, and a 0.001 learning rate. This results in a validation loss of 0.376 (or an RMSE of approximately 0.376 °C). In contrast, the optimal 3D-CNN uses 256 filters, a $5 \times 5 \times 5$ kernel size, no dropout, and a learning rate of 0.001. Its validation loss is 0.615 (RMSE of approximately 0.615 °C). ConvLSTM outperforms 3D-CNN in terms of validation loss, indicating that the LSTM recurrent mechanism more effectively captures temporal dependencies in this dataset. This suggests that the SST evolution pattern in southern Java waters has long-term temporal characteristics better captured by the LSTM memory mechanism than by the limited receptive field of the 3D convolution kernel.

In order to comprehend the contribution of each predictor variable to the model prediction, SHAP (SHapley Additive exPlanations)

analysis was performed on both models. Fig. 7 presents the distribution of SHAP values for all variables: (a) ConvLSTM and (b) 3D-CNN, where each point represents one prediction instance and the colour indicates the magnitude of the feature value.

In the ConvLSTM model Fig. 7 (a), chlorophyll-a (chl) exhibited the most significant impact, demonstrating a broad spectrum of SHAP values ranging from approximately -0.00020 to $+0.00010$. In this model, elevated chlorophyll levels (indicated by red dots) yielded negative SHAP values, which is consistent with the established physical relationship between upwelling and SST, where high chlorophyll levels are associated with lower SST. The zonal wind (u10) is the second most significant variable, exhibiting a bimodal distribution. Wind_speed and v10 demonstrate moderate contributions, while sss exhibits the smallest range. The ConvLSTM importance hierarchy is as follows: $\text{chl} > \text{u10} > \text{wind_speed} \approx \text{v10} > \text{sss}$. In contrast, the 3D-CNN model Fig. 7 (b) demonstrates a distinct hierarchical structure: $\text{chl} > \text{wind_speed} > \text{u10} \approx \text{v10} > \text{sss}$, where wind_speed is prioritised over u10. This suggests that 3D-CNN places greater reliance on direct wind magnitude to capture upwelling forcing.

The comparison demonstrates that ConvLSTM exhibits a broader spectrum of SHAP values, particularly for chl and u10, suggesting a more adaptable feature weighting based on temporal progression. As demonstrated by the 3D-CNN, a more concentrated distribution is exhibited, with more consistent feature weights being applied. However, this is at the expense of greater adaptability to temporal nuances. The findings of both models indicate that salinity variations are of minimal significance, thereby suggesting that variations in salinity may be redundant when considering the information provided by wind and chlorophyll. Table 4 explains the differences in the evaluation results between the best ConvLSTM and 3D-CNN models.

An evaluation of the test set reveals consistent performance disparities that are concomitant with validation loss. ConvLSTM achieved a mean root mean square error (RMSE) of 0.477°C and a mean absolute error (MAE) of 0.401°C , while 3D-CNN produced a mean RMSE of 0.675°C and a mean MAE of 0.585°C . ConvLSTM demonstrated an improvement in RMSE and MAE when compared to 3D-CNN. The pattern that emerges on a monthly basis indicates that ConvLSTM exhibit the most precise evaluation metrics for predictions made one month in advance, as compared to subsequent months. This may be due to the utilization of monthly data, which poses a challenge in predicting SST values in the context of the existing changes. To further elucidate the disparities in prediction accuracy, Fig. 8 presents a comparative analysis of ConvLSTM and 3D-CNN predictions alongside the actual SST values during the test data period.

Fig. 8 gives a visual comparison between the observed and predicted sea surface temperature (SST) offers further insight into the characteristics of both models' predictions. ConvLSTM predictions exhibit a smoother spatial pattern that aligns with actual observations, particularly during the period from July to September. This model effectively captures the SST gradient from north to south, maintaining consistent spatial coherence across the upwelling season. However, the model shows decreased accuracy in October–November, where predictions begin to deviate from observed patterns, particularly in the central and eastern regions. During November–December, ConvLSTM slightly underestimates the warming trend, though it still preserves the overall spatial structure better than 3D-CNN. The 3D-CNN prediction demonstrates a range of characteristics. From July through September, the prediction exhibits a greater degree of "homogeneity," characterized by diminished spatial variation relative to the observed data. This finding suggests that the model demonstrates a propensity for excessive spatial smoothing. Throughout the test period, 3D-CNN consistently shows reduced spatial detail, with predictions appearing more uniform across the domain. 3D-CNN exhibited a tendency to predict a more expansive yellow area (indicating high SST $\sim 28\text{--}29^\circ\text{C}$) in comparison to ConvLSTM. However, this prediction was of an exaggerated magnitude, as evidenced by the overestimation of the area depicted in brighter colors, particularly overestimating the extent of warm water masses in the eastern region.

According to the findings of this exhaustive evaluation, ConvLSTM has been demonstrated to be a more effective model for predicting SST one month in advance in the southern waters of Java. However, this advantage is accompanied by a longer training time and a more complex model architecture. This superiority was consistent across both the validation set and the test set, and manifested in its ability to capture spatial patterns with greater accuracy [37]. In operational applications where accuracy is paramount (e.g., for upwelling early warning systems or fisheries management), ConvLSTM emerges as the more reliable choice. Conversely, in scenarios where computational efficiency and inference speed are the primary constraints (e.g., real-time forecasting with limited resources), a 3D-CNN with a more suitable implementation can serve as a viable alternative, contingent upon the understanding of the accuracy trade-off that must be accepted.

Upwelling detection evaluation

The model's capacity to discern upwelling phenomena was assessed through an approach founded on the vertical thermal gradient between coastal and offshore regions. The detection methodology developed consists of three primary stages: the identification of coastal pixels, the calculation of thermal gradients, and the binary classification of upwelling.

The initial stage of the process entails the systematic identification of pixels that are located in coastal regions. For each longitude in the grid, the function identifies the northernmost sea pixel (the pixel closest to the coast of Java) as a representation of the coastal zone. To evaluate the sensitivity of upwelling detection to the coastal-offshore distance, three different pixel disparities were tested: 3 pixels ($\approx 75\text{ km}$), 6 pixels ($\approx 150\text{ km}$), and 9 pixels ($\approx 225\text{ km}$) to distinguish offshore pixels situated south from the coast. The process yielded 41 pairs of coastal-offshore points dispersed along the southern coast of Java.

The second stage employed a function to calculate the temperature difference between the coastal and offshore zones for each timestep and longitude at each of the three distances. The presence of negative coastal and offshore difference values indicates that the coastal zone is cooler than the offshore zone. This is a hallmark of active upwelling. Conversely, positive or zero values indicate non-upwelling conditions. This approach is predicated on the oceanographic principle that upwelling brings cold water masses to the surface in the coastal zone, thereby creating a clear thermal front with warmer offshore waters. The third stage implements binary

Table 5

Comparison results of upwelling detection between ConvLSTM and 3D-CNN.

Time	ConvLSTM			3D-CNN		
	3 Pixels	6 Pixels	9 Pixels	3 Pixels	6 Pixels	9 Pixels
July 2022	100 %	100 %	100 %	100 %	100 %	100 %
August 2022	100 %	100 %	100 %	100 %	100 %	100 %
September 2022	100 %	100 %	95.12 %	100 %	100 %	100 %
October 2022	92.68 %	95.12 %	97.56 %	95.12 %	100 %	97.56 %
November 2022	65.85 %	87.80 %	95.12 %	95.12 %	97.56 %	100 %
December 2022	95.12 %	100 %	100 %	100 %	100 %	100 %
Overall	92.28 %	97.15 %	97.97 %	97.87 %	99.59 %	99.59 %

classification with a simple threshold: upwelling is indicated if the difference is <0 , and no upwelling is indicated if it is greater than 0. This classification is applied to both actual data and model predictions at all three distance thresholds, enabling the evaluation of the model's capacity to detect upwelling across different spatial scales, in addition to the precision of absolute SST values. Table 5 illustrated a comparison of actual/true upwelling detections for the best ConvLSTM and 3D-CNN models across the three coastal-offshore distances.

The upwelling detection results reveal a clear sensitivity pattern to the coastal-offshore distance across both models. For the ConvLSTM model, detection accuracy increases progressively with distance: 92.28 % at 3 pixels, 97.15 % at 6 pixels, and 97.97 % at 9 pixels. This improvement indicates that ConvLSTM captures the thermal gradient more reliably at greater offshore distances, where the upwelling signal becomes more spatially distinct and less affected by nearshore noise or localized variability. Temporal analysis revealed that both ConvLSTM and 3D-CNN achieve perfect detection (100 %) during the peak upwelling months of July–August across all distances, demonstrating strong model performance when the upwelling signal is most pronounced. However, during the transition months of October–November, accuracy decreases, particularly at the 3-pixel distance where ConvLSTM achieves 92.68 % (October) and 65.85 % (November), improving to 95.12 % and 87.80 % at 6 pixels, and further to 97.56 % and 95.12 % at 9 pixels. This pattern suggests that the nearshore thermal gradient (3 pixels) is more variable and difficult to predict during transitional periods, while the broader offshore gradient (9 pixels) provides a more stable detection signal.

The 3D-CNN model demonstrates superior overall performance across all distances, achieving 97.87 % at 3 pixels, 99.59 % at 6 pixels, and 99.59 % at 9 pixels. Notably, 3D-CNN reaches near-perfect detection at both 6 and 9 pixels, with only minimal errors in the transition period. The temporal performance distribution exhibits distinct characteristics, exhibiting a pattern that is partially complementary to ConvLSTM, 3D-CNN maintains consistently high accuracy across all months, with perfect detection (100 %) in July, August, September (at 3 and 6 pixels), October (at 6 pixels), and November–December (at 9 pixels). The model shows minor degradation only at the 3-pixel distance during October–November (95.12 % and 95.12 %), which still exceeds ConvLSTM's performance at the same distance. The phenomenon can be explained by the different learning mechanisms of the two models.

The 3D-CNN model, with its concurrent 3D convolution strategy in the spatio-temporal dimension, exhibits heightened sensitivity to local patterns of spatial change within a constrained temporal window (6 months). Its ability to process spatial gradients across multiple scales simultaneously enables robust detection across all three distances, particularly excelling at the 6 and 9-pixel thresholds where the spatial structure of upwelling is more clearly defined. During periods of transition, 3D-CNN consistently maintains high accuracy across all distances, as it can directly capture changes in SST spatial gradients from 3D convolution without relying on temporal memory that might introduce bias from previous seasonal patterns. In contrast, ConvLSTM's recurrent memory mechanism, while advantageous during stable upwelling periods (July–September), introduces challenges during transitional months when SST patterns deviate from historical sequences.

Overall, the sensitivity analysis demonstrates that upwelling detection accuracy improves with increasing coastal-offshore distance for both models, with 9 pixels (≈ 225 km) providing the most robust detection performance. The 3D-CNN model emerges as the superior choice for upwelling detection applications, achieving near-perfect accuracy (99.59 %) at operational distances of 6–9 pixels, despite its lower performance in absolute SST prediction. This finding highlights an important distinction: models optimized for pixel-level SST accuracy (ConvLSTM) do not necessarily excel at spatial gradient-based detection tasks, while models that capture spatial patterns effectively (3D-CNN) can achieve superior performance in binary classification of oceanographic phenomena.

Validation of upwelling predictions against fisheries catch maps

Direct in situ validation of coastal upwelling intensity and structure is limited in the south of Java due to sparse long-term mooring and hydrographic observations. Therefore, an indirect but fishery-relevant validation approach was adopted by comparing the predicted upwelling patterns with independent fishing-effort density maps derived from Global Fishing Watch data. Upwelling enhances nutrient availability and primary productivity, which propagates through the food web and leads to aggregation of pelagic fish and fishing activity along upwelling filaments and frontal zones rather than within the coldest upwelling cores. Consequently, spatial fishing-effort patterns provide an applied proxy for the physical–biological coupling associated with upwelling dynamics. Comparing the space–time structure of predicted upwelling with fishing-effort maps thus evaluates whether the models capture the timing, location, and spatial organization of upwelling features that are relevant to ecosystem response and fishery behavior.

This validation does not rely on subjective interviews or anecdotal information from fishers; instead, it uses an independent,

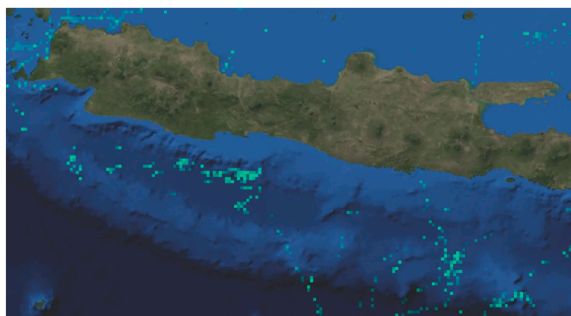


Fig. 9. Fishing Map July 2022.

satellite-derived dataset that reflects aggregated vessel behavior at basin scale. Such comparisons have been widely used in regions where biological response integrates physical forcing and where direct oceanographic observations are insufficient [32]. In addition, this approach can reveal lead–lag relationships between physical forcing and fishing activity, while also highlighting discrepancies caused by access limitations, fleet behavior, or reporting bias—factors that are not detectable using purely physical validation metrics. This validation strategy is particularly appropriate south of Java, where monsoonal forcing and ENSO variability modulate upwelling intensity and productivity, shaping pelagic fish distributions documented in previous ocean-color and fisheries studies [38–40].

In Fig. 9, as An example for July 2022 fishing-effort map that obtained from global fishing watch (<https://globalfishingwatch.org/>) show vessel concentration along the outer shelf and shelf-break south of Java ($\approx 50\text{--}150$ km offshore, $107\text{--}114^\circ\text{E}$). Three dominant clusters are evident: (1) a west–central band off Pangandaran–Cilacap ($\approx 107\text{--}110^\circ\text{E}$), (2) a central feature around $110\text{--}112^\circ\text{E}$, and (3) an eastern concentration off Blambangan–Banyuwangi ($\approx 112\text{--}114^\circ\text{E}$) extending toward the Bali Strait. The orientation and spacing of these clusters are consistent with expected upwelling corridors during the southeast monsoon, where fishing activity typically follows thermal and chlorophyll fronts rather than the coldest coastal core.

Both the 3D-CNN and ConvLSTM models reproduce a cool coastal band with an offshore warming gradient in July. The west–central frontal zone ($\approx 107\text{--}110^\circ\text{E}$), where ConvLSTM simulates a sharper cool tongue, aligns closely with the densest fishing cluster, indicating better representation of fishery-relevant frontal structure. In the eastern shelf ($\approx 112\text{--}114^\circ\text{E}$), fishing effort is strong; ConvLSTM captures cooling but places the front slightly too nearshore, while 3D-CNN produces a smoother and weaker signal, underestimating the intensity consistent with the observed effort hotspot. Overall, the spatial concordance between front-following fishing clusters and predicted upwelling patterns supports the ability of both models to resolve upwelling structures relevant to fisheries, with ConvLSTM showing superior coastal sharpness..

Limitations

In this research, we only used data from the South Sea of Java. Our analysis only compared two methods: ConvLSTM and 3D-CNN. Future research could explore other regions and incorporate additional methods.

Ethics statements

As an expert scientist and along with co-authors of the concerned field, the paper has been submitted with full responsibility, following due ethical procedure, and there are no duplicate publication, fraud, plagiarism, or concerns about animal or human experimentation.

Related research article

None

For a published article

None

CRediT author statement

Dwi Rantini: Supervision, Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing –review & editing, Software. **Rumaisa Kruba:** Conceptualization, Methodology, Validation. **Yudi Haditjar:** Supervision, Conceptualization, Methodology, Validation. **Muhammad Ikhwan:** Supervision, Conceptualization, Methodology, Validation. **Yusuf Jati Wijaya:** Supervision, Conceptualization, Methodology, Validation. **Aris Ismanto:** Supervision, Conceptualization, Methodology, Validation. **Muhammad Mahdy Yandra:** Conceptualization, Methodology, Writing – original draft, Writing –review & editing,

Validation. **Hafiz Rahman:** Conceptualization, Methodology, Writing – original draft, Writing –review & editing, Validation. **Arip Ramadan:** Writing – original draft, Writing –review & editing. **Fazidah Othman:** Writing – original draft, Writing –review & editing

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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